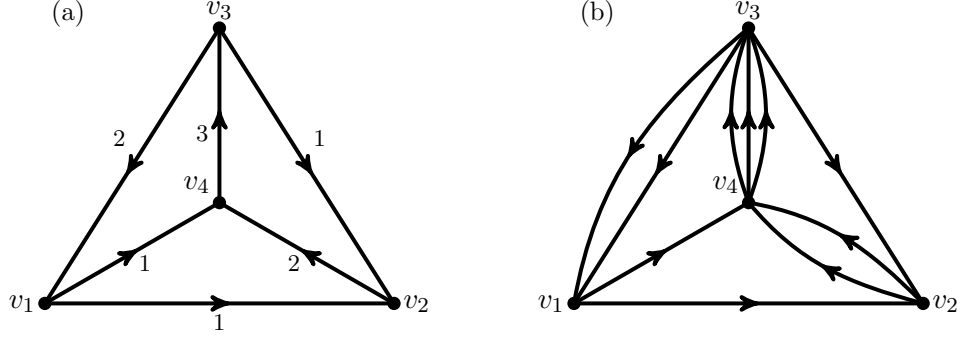


FIGURE 1. An example for K_4 .

from v_1 . Initially we may let P follow either $\mathbf{e}_{14}$ or $\mathbf{e}_{12}$. Choose to follow $\mathbf{e}_{14}$. Keep extending P in this way, we may get $P = \mathbf{e}_{14}\mathbf{e}_{43}\mathbf{e}_{31}\mathbf{e}_{12}\mathbf{e}_{24}\mathbf{e}_{43}\mathbf{e}_{31}$. At this stage P is a circuit and the “while” loop is exited, and we have to translate P to find a “non-saturated” vertex. The first such a vertex we get is v_4 . By translation, now P becomes $\mathbf{e}_{43}\mathbf{e}_{31}\mathbf{e}_{12}\mathbf{e}_{24}\mathbf{e}_{43}\mathbf{e}_{31}\mathbf{e}_{14}$. Extending P , we will finally get a direction-consistent circuit $C = \mathbf{e}_{43}\mathbf{e}_{31}\mathbf{e}_{12}\mathbf{e}_{24}\mathbf{e}_{43}\mathbf{e}_{31}\mathbf{e}_{14}\mathbf{e}_{43}\mathbf{e}_{32}\mathbf{e}_{24}$ such that $C^{\text{ab}} = \alpha$.

3.2. Counting direction-consistent circuits: generalized BEST theorem.

The well-known BEST theorem, named after de Bruijn and van Aardenne-Ehrenfest [vAEdB51] and Smith and Tutte [TS41], provides an efficient algorithm of counting Eulerian cycles on an Eulerian digraph $G_{\mathfrak{o}}$ (meaning that the in-degree and out-degree at each vertex of $G_{\mathfrak{o}}$ are identical).

Recall that an arborescence T of $G_{\mathfrak{o}}$ with a root vertex w is a spanning tree of G such that all edges on T are oriented towards w with respect to $\mathfrak{o}$. Recall also that the Laplacian matrix of $G_{\mathfrak{o}}$, denoted $L(G_{\mathfrak{o}})$, is defined as: (i) for $i \neq j$, $(L(G_{\mathfrak{o}}))_{ij}$ is the negation of the number of positively oriented edges with respect to $\mathfrak{o}$ with initial vertex v_i and terminal vertex v_j , and (ii) $(L(G_{\mathfrak{o}}))_{ii} = -\sum_{j \neq i} (L(G_{\mathfrak{o}}))_{ij}$. Removing the row and column of $L(G_{\mathfrak{o}})$ with respect to w , we get the reduced Laplacian $L^{\text{red}(w)}(G_{\mathfrak{o}})$ at w .

Denote by $\text{ec}(G, \mathfrak{o})$ the number of Eulerian cycles on $G_{\mathfrak{o}}$, and $\text{In}_w(G, \mathfrak{o})$ the number of arborescences $G_{\mathfrak{o}}$ rooted at w . Then the BEST theorem states that $\text{ec}(G, \mathfrak{o}) = \text{In}_w(G, \mathfrak{o}) \cdot \prod_{v \in V(G)} (\deg_{\mathfrak{o}}^+(v) - 1)!$ where $\deg_{\mathfrak{o}}^+(v)$ is the out-degree of $G_{\mathfrak{o}}$ at $v \in V(G)$. Note that by a directed version of Kirchhoff’s matrix-tree theorem, $\text{In}_w(G, \mathfrak{o})$ is actually independent of w , and can be efficiently computed as the determinant of the reduced Laplacian matrix $L^{\text{red}(w)}(G_{\mathfrak{o}})$.

Before making a generalization of the BEST theorem, we first introduce a notion of weighted Laplacian here.

Definition 3.5. Consider a graph G . Let $\alpha \in C_1(G, \mathbb{Z})$ be universal. Let $c_{\mathbf{e}} = \deg_{\alpha}(\mathbf{e})$ for all $\mathbf{e} \in \mathbf{E}(G)$. Then $\alpha = \sum_{\mathbf{e} \in \mathbf{E}_{\mathfrak{o}(\alpha)}(G)} c_{\mathbf{e}} \cdot \mathbf{e}$ with all $c_{\mathbf{e}} > 0$. The *weighted Laplacian* $L_{\alpha}(G)$ of G with respect to α is defined as: $(L_{\alpha}(G))_{ij} = -\sum_{\mathbf{e} \in \mathbf{E}_{\mathfrak{o}(\alpha)}(G), \mathbf{e}(0)=v_i, \mathbf{e}(1)=v_j} c_{\mathbf{e}}$ for all $i \neq j$, and $(L_{\alpha}(G))_{ii} = \sum_{\mathbf{e} \in \mathbf{E}_{\mathfrak{o}(\alpha)}(G), \mathbf{e}(0)=v_i, \mathbf{e}(1) \neq v_i} c_{\mathbf{e}}$ for all diagonal entries. Removing the row and column of L_{α} with respect to w , we get the *reduced weighted Laplacian* $L_{\alpha}^{\text{red}(w)}(G)$.